AUTOMOTIVE SOFTWARE ARCHITECTURE

Synthetic HLD Specification for RAG Retrieval, Grounding & Citation Testing

Document ID: SYN-HLD-042 **Version:** 1.2 **Status:** Test Specification

Important: This is fictional test material for evaluating a document-intelligence application. It is not an official AUTOSAR specification and must not be used as an engineering reference.

Purpose

Use this PDF to test upload, extraction, chunking, semantic retrieval, grounded answers, source-page citations, entity extraction, validation findings, and comparison. It includes specific facts and intentional ambiguities.

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1. System Overview

The fictional Aurora Vehicle Control Platform (AVCP) coordinates torque requests, battery monitoring, and vehicle diagnostics. It contains four software components: TorqueCoordinator (TC), BatterySupervisor (BS), DiagnosticGateway (DG), and EventRecorder (ER).

All values in this document are invented test values, not safety recommendations. The reference system uses a 12 V auxiliary network and a 400 V traction battery.

1.1 Operating Modes

AVCP defines Startup, Active, and Degraded modes. Startup ends after BatterySupervisor publishes a valid BatteryStatus and TorqueCoordinator receives the first valid TorqueRequest. In Active mode, TorqueCoordinator accepts requests up to 320 Nm. In Degraded mode, the configured torque limit is 120 Nm.

1.2 Data Ownership

BatterySupervisor is the sole owner of batteryStateOfCharge and batteryTemperature. TorqueCoordinator owns requestedTorqueNm and grantedTorqueNm. DiagnosticGateway owns activeDiagnosticCode. EventRecorder stores timestamped event records but must not modify values owned by other components.

1.3 Deployment

The reference deployment uses a 2-core controller, 256 MB RAM for the application partition, and a 10 ms base scheduling cycle. EventRecorder writes batches to non-volatile storage every 500 ms. Its maximum in-memory queue is 2,000 records.

2. Components and Responsibilities

Component	Responsibility	Owned data / outputs
TorqueCoordinator (TC)	Validates torque requests and applies mode limits.	grantedTorqueNm; torqueLimitStatus
BatterySupervisor (BS)	Monitors battery state of charge and temperature.	batteryStateOfCharge; batteryTemperature; BatteryStatus
DiagnosticGateway (DG)	Maps detected faults to diagnostic codes.	activeDiagnosticCode; DiagnosticEvent
EventRecorder (ER)	Persists timestamped events and retrieves them by event ID.	EventRecord; oldestRetainedEventId

2.1 TorqueCoordinator

TorqueCoordinator receives TorqueRequest from the vehicle request adapter. It rejects negative values and values above 400 Nm. Valid requests are clamped to the active mode limit: 320 Nm in Active mode or 120 Nm in Degraded mode. It publishes grantedTorqueNm every 10 ms.

2.2 BatterySupervisor

BatterySupervisor samples battery temperature every 100 ms and state of charge every 250 ms. It sets batteryWarning when temperature is greater than 58 °C or state of charge is below 15%. It publishes BatteryStatus every 100 ms.

2.3 DiagnosticGateway and EventRecorder

DiagnosticGateway maps fault F-17 to DTC-0x17 and F-29 to DTC-0x29. EventRecorder assigns monotonically increasing event IDs. If its queue reaches 2,000 records, it must raise QueueCapacityWarning before accepting more records.

3. Interfaces and Signals

Interface	Provider → Consumer	Payload / constraints
TorqueRequest	VehicleRequestAdapter → TC	requestedTorqueNm: signed 16-bit integer; valid input 0–400 Nm
BatteryStatus	BS → TC, DG	batteryStateOfCharge: 0–100%; batteryTemperature: -40–125 °C; warning flag
DiagnosticEvent	DG → ER	diagnosticCode, sourceFaultCode, timestampMs
TorqueStatus	TC → ER, DG	grantedTorqueNm: 0–320 Nm Active; 0–120 Nm Degraded
EventQuery	ServiceTool → ER	eventId: unsigned 32-bit integer; returns EventRecord or NotFound

3.1 Signal Details

requestedTorqueNm is validated before grantedTorqueNm is calculated. torqueLimitStatus uses NORMAL, LIMITED, or REJECTED.

batteryStateOfCharge is an integer percentage and batteryTemperature is measured in degrees Celsius. Both are produced by BatterySupervisor; no other component may write them.

activeDiagnosticCode is updated by DiagnosticGateway. A cleared fault is represented by NONE. DiagnosticGateway sends an event to EventRecorder when the active code changes.

3.2 Communication Constraints

All inter-component messages include a timestamp in milliseconds since controller startup. A message older than 100 ms is stale for TorqueCoordinator and must not be used for a new torque decision. EventQuery must return within 50 ms when the event exists in the in-memory index.

4. Timing and Resource Requirements

Operation	Required period / limit
Base controller scheduling cycle	10 ms
TorqueCoordinator publication	10 ms
Battery temperature sampling	100 ms
Battery state-of-charge sampling	250 ms
EventRecorder batch persistence	500 ms
Maximum EventRecorder queue	2,000 records
Indexed EventQuery response	≤ 50 ms
Stale message threshold	100 ms

4.1 Memory Budget

The application partition has a 256 MB RAM budget. EventRecorder is allocated up to 32 MB for its event queue and indexing structures. The remaining application memory must not exceed 224 MB under the reference load.

4.2 Timing Interpretation

A periodic task must not be scheduled faster than its stated period unless another requirement explicitly permits it. A maximum response-time requirement means the measured response must be at or below that limit.

5. Validation Rules

VAL-01 — Torque input range: Reject requestedTorqueNm below 0 Nm or above 400 Nm.

VAL-02 — Mode torque limit: grantedTorqueNm must not exceed 320 Nm in Active mode or 120 Nm in Degraded mode.

VAL-03 — State-of-charge bounds: batteryStateOfCharge must be between 0% and 100%, inclusive.

VAL-04 — Temperature bounds: batteryTemperature must be between -40 °C and 125 °C, inclusive.

VAL-05 — Stale messages: TorqueCoordinator must not use data older than 100 ms for a new torque decision.

VAL-06 — Queue capacity: EventRecorder must raise QueueCapacityWarning when the queue reaches 2,000 records and must not silently overwrite unpersisted events.

VAL-07 — Ownership: Only BatterySupervisor may write batteryStateOfCharge and batteryTemperature.

VAL-08 — Diagnostic mapping: F-17 maps to DTC-0x17; F-29 maps to DTC-0x29.

6. Intentional Test Issues

These statements are deliberately incomplete or ambiguous. A good RAG system should retrieve them accurately and identify ambiguity instead of inventing a resolution.

ISSUE-01 — Torque boundary: Section 2.1 rejects values above 400 Nm, while the interface lists the valid range as 0–400 Nm. Test inclusive/exclusive boundary handling.

ISSUE-02 — Missing timestamp: The stale-message rule does not define behavior when a message timestamp is missing.

ISSUE-03 — Queue boundary: The queue warning threshold is 2,000, but the exact accept/reject behavior for the 2,000th record is not fully defined.

ISSUE-04 — Battery warning: The document does not state whether batteryWarning forces Degraded mode.

ISSUE-05 — Timestamp synchronization: The document does not define timestamp synchronization across separately restarted components.

7. Glossary and Revision Notes

Term	Meaning
AVCP	Aurora Vehicle Control Platform
DTC	Diagnostic Trouble Code
HLD	High-Level Design
RAG	Retrieval-Augmented Generation
TC / BS / DG / ER	TorqueCoordinator / BatterySupervisor / DiagnosticGateway / EventRecorder

Revision 1.2 adds data ownership, message-age constraints, and validation cases for retrieval and citation testing.

End of synthetic test specification.